BEHAVIORAL FINANCE AND ANALYST FORECASTS

Learning Outcome

- explain how behavioral factors affect analyst forecasts and recommend remedial actions for analyst biases

Studies have shown that experts in many fields persistently make forecasting errors arising from behavioral biases, and investment analysts' models of financial statements are in no way immune. To improve forecasts and the investment decisions based on them, analysts must be aware of the impact of biases and potential remedies for them. Five key behavioral biases that influence analyst forecasts are overconfidence, illusion of control, conservatism, representativeness, and confirmation bias.

Overconfidence in Forecasting

Overconfidence bias occurs when people demonstrate unwarranted faith in their own abilities. Studies have identified that 90 percent *confidence intervals* for forecasts, which should leave only 10 percent error rates, turn out to be wrong as much as 40 percent of the time (**Russo and Schoemaker 1992**). Studies have also suggested that individuals are more confident when making contrarian predictions that counter the consensus. That is, overconfidence arises more frequently when forecasting what others do not expect (**Dunning, Griffin, Milojkovic, and Ross 1990**).

To mitigate overconfidence bias, analysts should record and share their forecasts and review them regularly, identifying *both* the correct and incorrect forecasts they have made. Given the wide range of outcomes for most financial variables, an analyst will likely find that they have been wrong as much or more often than they have been right. The goal is to recognize that forecast error rates are high, so mitigating actions that widen the confidence interval of forecasts should be taken. One such action is **scenario analysis**. By asking, "Where could I be wrong and by how much?," an analyst can generate different forecast scenarios.

EXAMPLE 1

Mitigating Overconfidence: Scenario Analysis for Rémy

In the prior lesson, a financial statement model was constructed for Rémy Cointreau Group that includes only one set of forecasted numbers, or one scenario. Creating several more scenarios is an important modeling step because the range of outcomes for the most important variable is wider than a single point.

Three important variables in the forecast of free cash flow are organic sales growth in the Cognac segment, EBIT margin, and net working capital as a percentage of sales. A benefit of the spreadsheet-driven model is that the forecasts can be easily modified to calculate different free cash flow estimates. The base case inputs and forecast for 2024E free cash flow to the firm, as well as figures for two different scenarios, are shown in Exhibit 15.

Alternative Scenario 1 assumes that the Cognac segment's organic growth remains the same as its FY2021 rate, an EBIT margin of 23.6 percent, where it was before the COVID-19 pandemic, and working capital of 86 percent of sales, also the pre-pandemic level from FY2019. Alternative Scenario 2 assumes the same Cognac segment organic growth rate as the base case but an EBIT margin of 25.0 percent and working capital of 90 percent of sales. This scenario reflects strong growth but a

As Exhibit 15 demonstrates, there is a wide range of free cash flow estimates for 2022E–2024E because of a wide range of reasonable inputs for key variables.

Exhibit 15:

Calculating Free Cash Flow to the Firm as Basis for a DCF Valuation Model (euro millions)

Base Case	2022E	2023E	2024E
Cognac segment organic growth	11.3%	10.2%	10.2%
EBIT margin	24.8%	26.2%	27.6%
Working capital as percent of sales	91%	78%	65%
Free cash flow to the firm est.	230	267	300
Alternative Scenario 1	**2022E**	**2023E**	**2024E**
Cognac segment organic growth	4.0%	4.0%	4.0%
EBIT margin	23.6%	23.6%	23.6%
Working capital as percent of sales	86%	86%	86%
Free cash flow to the firm est.	318	133	129
Alternative Scenario 2	**2022E**	**2023E**	**2024E**
Cognac segment organic growth	11.3%	10.2%	10.2%
EBIT margin	25.0%	25.0%	25.0%
Working capital as percent of sales	90%	90%	90%
Free cash flow to the firm est.	240	109	105

Illusion of Control

A bias often linked to overconfidence, illusion of control is a tendency to overestimate the ability to control what cannot be controlled and to take ultimately fruitless actions in pursuit of control. This bias often manifests in analysts' beliefs that forecasts can be rendered more accurate in two ways: by acquiring more information and opinions from experts and by creating more granular and complex models. Although additional information and complexity in model specification can improve forecasting accuracy, there are diminishing marginal returns. The amount of material information available for an investment is finite and adding immaterial information will mislead. Complex models tend to be overfitted to historical data sets which do not prove robust in a range of environments that include never-before-seen outliers. Excessive breadth of data and model complexity can also conceal assumptions and make updating forecasts upon the receipt of new information difficult. Finally, analysts face significant opportunity costs; additional hours modeling one company could mean that the analyst will examine fewer opportunities in total.

Beyond awareness of the bias and the recognition that uncertainty is an inherent characteristic in investments, illusion of control can be mitigated by restricting modeling variables to those that are regularly disclosed by the company, focusing on the most important or impactful variables, and speaking only with those who are likely to have unique or significant perspectives.

EXAMPLE 2

Illusion of Control: How Much Model Complexity?

Rémy Cointreau Group regularly reports revenues by segment and by geographic region (Europe/Middle East/Africa, the Americas, and Asia Pacific). It does not disclose segment revenue by geographic region (e.g., the Cognac segment revenue in the Asia Pacific region), nor does it disclose revenue by sales channel, such as

rates in specific regions and growth rates of specific channels, even though the actual numbers are not disclosed. Such a practice is common, especially during the COVID-19 pandemic because large sales channel shifts occurred: travel retail in most regions experienced declines >90 percent, sales shifted from bars and restaurants to retailers for at-home consumption, and different geographies were affected by the pandemic at different times.

An analyst might be tempted to collect all these growth rates and other anecdotal figures that management discloses on its earnings calls and, perhaps by combining them with third-party estimates of sales, to build an extensive revenue model for Rémy in which each segment is broken out into geographic regions and sales channels.

Although such an endeavor might be useful to set expectations and to monitor over time, building the revenue forecast in this way would introduce several problems and probably not materially improve accuracy. First, because the data used in the model are not regularly disclosed, there is no way to check actuals versus estimates. Second, model construction would take dozens of hours. Finally, and perhaps most importantly, whether the constituent small parts of such a model would be accurate is unclear, which would not make the consolidated revenue forecast any more accurate than a simpler model.

Conservatism Bias

Conservatism bias is a bias in which people maintain their prior views or forecasts by inadequately incorporating new information. This often happens in forecasting when an analyst does not update their forecasting after receiving conflicting information, such as disappointing earnings results or a competitor action. Although the most common form of conservatism is the reluctance to incorporate new negative information into a forecast, analysts could also fail to adequately incorporate positive information and thus have estimates that are too low. A different name for conservatism bias in this context is anchoring and adjustment, referring to an analyst using their prior estimates as an “anchor” that is subsequently adjusted. Although nothing is wrong with modifying a previous forecast, the previous forecast or anchor tends to exert significant influence; in other words, the adjustment is too small, and the updated forecast is too close to the previous forecast.

Conservatism bias can be mitigated by reviews of forecasts and models by an investment team at a regular interval, such as each quarter, and by creating flexible models with fewer variables, to make changing assumptions easier. Because conservatism bias is related to overconfidence and the illusion of control, mitigating those biases can also serve to mitigate conservatism.

EXAMPLE 3

Conservatism Bias: Rémy Management Guidance for FY2022

The base case forecasts in the Rémy Cointreau Group model call for organic revenue growth of 11.3 percent and net income growth of 24 percent in FY2022E over FY2021. However, during the earnings call for the fourth quarter of FY2021, Rémy management gave the following guidance for FY2022:

- “Fiscal year 2022 will be a strong year of growth and investment, and we are on track to achieve our 2030 [objectives of a 72% gross margin and 33% operating margin].”
- “Being ahead of [our] 2030 strategic plan and given the favorable environment, [we] have decided to revise up [our] strategic investments [in sales and marketing]

- Fiscal year 2022 will have "top-line and bottom-line growth in the mid-teens in organic terms."

Based on these comments, your colleague suggests revising the Rémy model slightly by reducing the operating margin forecast to reduce net income growth from 24 percent to 20 percent.

1. What behavioral bias does your colleague's suggestion exhibit, and what research or steps should be taken, if any, with respect to revising the Rémy model? Explain your answer.

   Show Solution

Representativeness Bias

Representativeness bias refers to the tendency to classify information based on past experiences and known classifications. New information might resemble or seem representative of familiar elements already classified but can in fact be very different and is better viewed from a different perspective. In these instances, the classification reflex can deceive, producing an incorrect understanding that biases all future thinking about the information. Base-rate neglect is a common form of representativeness bias in forecasting. In base-rate neglect, a phenomenon's rate of incidence in a larger population, or characteristics of a larger class to which a specific member belongs—its base rate—is neglected in favor of situation- or member-specific information. Considering the base rate is sometimes known as the "outside view," while the situation-specific is known as the "inside view."

For example, an analyst is modeling operating costs and margins for a biopharmaceutical company. The "inside view" approach would consider company-specific factors such as the types of drugs the company sells, the number of salespeople needed in each geographic region for each drug, and so on. The "outside view" approach would view the company as a member of the "biopharmaceuticals" industry, of which there are many others, and use industry or sector averages for gross margin, R&D expense as percentage of sales, and so on in the model.

Neither the outside nor inside view is superior; what makes for a superior forecast is considering both. One way of doing so is by starting with the base rate but determining which factors make the target company different from the base rate or class average and what the implications of those differences are, if any. For example, the analyst modeling the biopharmaceuticals company might start with industry averages in the model but change some of the variables to account for factors such as royalties versus product sales revenues, geographic composition of revenues, and whether the company is likely to face patent expirations on its products over the forecast period.

EXAMPLE 4

Considering Base Rates for Rémy

While constructing the Rémy model in the prior lesson, little attention was given to comparable companies or to the broader industry to which Rémy belongs. In other words, the model was constructed primarily with the "inside view." In this example, Rémy is put in the context of six other spirits-focused alcohol companies: Brown-Forman Corporation, Pernod Ricard SA, Davide Campari-Milano N.V., Diageo plc, Becle S.A.B de C.V. (Cuervo), and the Wine & Spirits segment of LVMH (LVMH W&S) for the five most recently reported fiscal years at the time of analysis. The variable used for the industry comparison is the five-year average of EBIT margin because it is a key model input, and the profitability of an individual company is strongly influenced by industry profitability. Many of these peer companies are significantly larger by revenue than Rémy, which is useful because we have modeled

Exhibit 16:

EBIT Margin Comparison of Spirits Companies, Last Five Reported Fiscal Years (euro millions)

EBIT margin	MRY-4	MRY-3	MRY-2	MRY-1	Most Recent Year (MRY)
Rémy	20%	20%	24%	19%	23%
Brown Forman	34%	32%	34%	32%	34%
Pernod	24%	25%	26%	26%	12%
Campari	22%	26%	25%	25%	17%
Diageo	27%	30%	30%	31%	18%
Cuervo	23%	26%	20%	18%	20%
LVMH W&S	31%	31%	32%	31%	29%
Peer average (ex Rémy)	27%	28%	28%	27%	22%
Peer five-year average (ex Rémy)	26%				

1. Evaluate the base case forecasts in the Rémy model as well as Rémy's management's FY2030 objective of a 33 percent operating margin considering the analysis in Exhibit 16.

   Show Solution

Confirmation Bias

Confirmation bias is the tendency to look for and notice what confirms prior beliefs and to ignore or undervalue whatever contradicts them. A common manifestation of this bias among investment analysts is to structure the research process in pursuit of only positive news or certain criteria, or with a narrow scope. For example, an analyst might research a particular company but conduct only cursory research on its competitors and companies that offer substitute products. An analyst who has a positive view on a company might speak only to other analysts who share that view and the company's management, all of whom will likely tell the analyst what they want to hear and already know. Confirmation bias is closely related to overconfidence and representativeness biases.

The extent to which company management can be excessively optimistic is shown in Exhibit 17, which analyzes the annual report of a major European bank for 2007, published mere months before it entered bankruptcy and was nationalized.

Speaking with management is valuable given their role and should not be excluded from the research process, but analysts must be aware of management's inherent bias and seek differing perspectives, especially when examining a company with significant controversy. Two approaches to mitigating confirmation bias in the forecasting process are to speak to or read research from analysts with a negative opinion on the security under scrutiny and to seek perspectives from colleagues who are not economically or psychologically invested in the subject security.

EXAMPLE 5

Management Optimism

Consider this text analysis of the chairman's statement and business review in the 2007 annual report of a major European bank published in 2008, a few months before the bank was rescued by the government.

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Disappoint/disappointed	0	Good	55
Bad/badly	0	Excellent	12
Poor	0	Success/successful	35
Weaker/weakening	7	Improvement	23
Slowdown	6	Strong/stronger/strongly	78

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